

Predicting NBA All-Star Selection from Regular-Season Performance

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1. Problem Statement

Each February, the NBA selects approximately 24 players for the All-Star Game, one of the most-watched events in professional basketball. Selection is determined by a combination of fan voting (50%), media ballots (25%), and player voting (25%) for starters, while coaches select the reserves. For sports media companies, anticipating which players will be selected is valuable: it drives pre-event coverage decisions, prediction content, and audience engagement strategies.

Prediction question: Can we predict whether an NBA player will be selected as an All-Star based on their regular-season per-game statistics?

Stakeholder: A sports media company (e.g., ESPN, The Ringer) deciding which borderline players to feature in pre-All-Star coverage and prediction articles. Accurate predictions would allow editors to assign feature stories, schedule interviews, and create interactive content weeks before the official announcement, improving audience engagement and advertising revenue.

Decision enabled: Given a player's mid-season stat line, should the media company invest editorial resources in covering this player as a likely All-Star? A predicted probability above 50% triggers coverage allocation.

Prediction vs. causation: This analysis is strictly predictive. We use statistical associations between per-game stats and All-Star selection to classify players. We make no causal claims. A high coefficient on points per game does not mean that scoring more points causes selection; it means PTS is a useful signal. Market size, team record, narrative, and popularity all influence selection but are not captured in our feature set.

2. Data Description

Source: The dataset comes from the Kaggle NBA/ABA/BAA Stats repository (Sumitro Datta, accessed April 19, 2026), which compiles historical statistics originally sourced from Basketball Reference. We use the 'Player Per Game' table for regular-season statistics and the 'All-Star Selections' table for labels.

Scope: 10 NBA seasons from 2014-15 through 2023-24. After filtering to players with at least 20 games played (to exclude 10-day contracts and injured players) and deduplicating traded players (keeping the season-total row), the final dataset contains 4,255 player-season observations.

Attribute	Value
Observations	4,255 player-seasons
Seasons	10 (2014-15 to 2023-24)
Features	13 per-game statistics
Target	all_star (binary: 1 = selected, 0 = not)
Class balance	6.3% All-Star (266 of 4,255)
Source	Kaggle NBA/ABA/BAA Stats (Sumitro Datta)

Table 1: Dataset overview.

Features: Points (PTS), assists (AST), total rebounds (TRB), steals (STL), blocks (BLK), minutes played (MP), field goal percentage (FG%), three-point percentage (3P%), free throw percentage (FT%), turnovers (TOV), age, games played (G), and games started (GS). All statistics are per-game averages.

Data quality: Shooting percentages (3P%, FT%) contain missing values when a player has zero attempts in that category. This is structurally missing (MCAR) and imputed with 0. Games started (GS) has occasional missing values, also imputed with 0. No other missingness was observed. Traded players appear in multiple rows; we retain only the season-total (TOT) row to avoid double-counting.

3. Methodology

We compare three classification models, each addressing the severe class imbalance (6.3% positive rate) through different mechanisms.

Train-test split

The dataset is split 80/20 with stratification on the target variable and `random_state=42` for reproducibility. The training set contains 3,404 observations; the test set contains 851. All hyperparameter tuning and cross-validation are performed exclusively on the training set.

Model 1: Logistic Regression

A regularized logistic regression with `class_weight='balanced'` to upweight the minority class. Features are standardized using `StandardScaler` (fit on training data only). The regularization parameter `C` is tuned via 5-fold stratified cross-validation over `{0.01, 0.1, 1, 10, 100}`, optimizing F1 score.

Model 2: Random Forest

An ensemble of decision trees with `class_weight='balanced'`. Hyperparameters tuned via `GridSearchCV`: `n_estimators` in `{100, 200}`, `max_depth` in `{5, 10, None}`, and `min_samples_leaf` in `{2, 5}`. No feature scaling required.

Model 3: Gradient Boosting

A sequential boosting ensemble. Hyperparameters tuned via `GridSearchCV`: `n_estimators` in `{100, 200}`, `max_depth` in `{3, 5}`, `learning_rate` in `{0.1}`, and `min_samples_leaf` in `{5}`. No feature scaling required.

Evaluation

All models are evaluated on the held-out test set using accuracy, precision, recall, F1 score, and AUC-ROC. We report 95% bootstrap confidence intervals (1,000 resamples) for each metric. We select the best model by F1 score, which balances precision and recall in the presence of class imbalance.

4. Results

Model	Accuracy	Precision	Recall	F1	AUC-ROC
Logistic Regression	0.946	0.539	0.906	0.676	0.984
Random Forest	0.962	0.691	0.717	0.704	0.977
Gradient Boosting	0.959	0.737	0.528	0.615	0.977

Table 2: Test set performance. Random Forest (highlighted) achieves the best F1.

Model	F1 [95% CI]	Precision [95% CI]	Recall [95% CI]	AUC-ROC [95% CI]
Logistic Reg.	0.676 [0.580, 0.755]	0.539 [0.437, 0.636]	0.906 [0.812, 0.979]	0.984 [0.976, 0.992]
Random Forest	0.704 [0.595, 0.795]	0.691 [0.569, 0.813]	0.717 [0.595, 0.833]	0.977 [0.964, 0.987]
Grad. Boost.	0.615 [0.500, 0.727]	0.737 [0.600, 0.872]	0.528 [0.396, 0.667]	0.977 [0.964, 0.987]

Table 3: 95% bootstrap confidence intervals (1,000 resamples).

Key findings: The Random Forest model achieves the highest F1 score (0.704), offering the best balance between precision (69.1%) and recall (71.7%). Logistic Regression achieves the highest AUC-ROC (0.984) and recall (90.6%) but sacrifices precision (53.9%), producing many false positives. Gradient Boosting achieves the highest precision (73.7%) but misses nearly half of actual All-Stars (recall = 52.8%).

Feature importance: Across all three models, points per game (PTS), minutes played (MP), and assists (AST) emerge as the top predictive features. This aligns with domain intuition: All-Stars are typically high-scoring starters who play heavy minutes. Field goal percentage and games started also contribute meaningfully. Turnovers (TOV) carry positive predictive weight, likely because high-usage stars handle the ball more frequently.

Predictive importance, not causal effect. These rankings reflect which stats are most useful for prediction, not which stats cause All-Star selection.

5. Recommendation

Recommendation: We recommend the Random Forest model ($F1 = 0.704$, 95% CI [0.595, 0.795]) as the production model for identifying likely All-Stars. At the default 50% probability threshold, it correctly identifies approximately 72% of actual All-Stars while maintaining 69% precision, meaning about 7 in 10 players flagged as likely All-Stars are actually selected.

Actionable use: A sports media editor should use the model's predicted probabilities to prioritize coverage. Players with predicted probabilities above 70% warrant feature articles and dedicated coverage. Players in the 50-70% range are borderline candidates worth monitoring. Players below 50% are unlikely selections and should not receive All-Star-focused editorial investment.

Uncertainty: The model's confidence intervals are wide ($F1$ CI spans 0.595 to 0.795), reflecting genuine uncertainty in a small positive class (266 All-Star observations across 10 seasons). The model should be used as one input among many, not as the sole decision criterion.

Limitations

First, the model uses only box-score statistics and cannot capture fan popularity, media narrative, team market size, or team record, all of which influence selection. Second, the class imbalance (6.3% positive rate) means even small changes in the threshold substantially affect precision-recall tradeoffs. Third, the model is trained on a 10-season window; changes in NBA style (e.g., increased three-point shooting) may cause distribution shift over time. Fourth, replacement All-Stars (injury substitutes) are included in our labels, which may introduce noise.

Next steps

Future work should incorporate additional features such as team win-loss record, player popularity metrics (social media following, jersey sales), conference affiliation, and advanced stats (PER, Win Shares, VORP). A time-aware split (training on earlier seasons, testing on later ones) would better simulate real-world deployment. Finally, calibrating predicted probabilities using Platt scaling or isotonic regression could improve the reliability of probability estimates for threshold-based editorial decisions.